$$E_1 = \frac{7}{4}\frac{ml^2}{T^2} \qquad E_2 = \frac{1}{4}\frac{ml^2}{T^2} \qquad E_3 = \frac{17}{36}\frac{ml^2}{T^2} \tag{9}$$

$t = T/2$, C_1 at $u = 0$: all speeds again pass C_1. Those with speed v_n' satisfying

$$\frac{l}{v_n'} = \frac{n}{2}T \qquad n = 1,2,\cdots \tag{10}$$

can also pass C_2. Their speeds are

$$v_n' = \frac{2l}{nT} \qquad n = 1,2,\cdots \tag{11}$$

Times from C_1 to C_2 and C_2 to G, respectively:

$$\Delta t_{n1}' = \frac{n}{2}T \qquad \Delta t_{n2}' = \frac{n}{4}T \tag{12}$$

Time of arrival at G:

$$t_n' = \frac{T}{2} + \Delta t_{n1}' + \Delta t_{n2}' = \frac{3n+2}{4}T \tag{13}$$

At $t = 10T/4$, D_2 shuts, hence

$$t_n' \leqslant \frac{10}{4}T \tag{14}$$

By (13) and (14), n being integral,

$$n_{max} = 2 \tag{15}$$

These particles enter the field with energies

$$E_n' = \frac{1}{2}mv_2'^2 + E_0\sin\left[\left(\frac{2\pi}{T}\frac{3n+2}{4}\right)T + \frac{\pi}{4}\right]$$
$$n = 1,2 \tag{16}$$

Entry energies are, respectively,

$$E_1' = \frac{9}{4}\frac{ml^2}{T^2} \qquad E_2' = \frac{3}{4}\frac{ml^2}{T^2} \tag{17}$$

At $t = 3T/4$, D_1 shuts: no later entry to C_1. By (9), (17), max entry energy is E_1', giving greatest speed V_1'. Using $E_1' = \frac{1}{2}MV_1'^2$,

$$V_1' = \sqrt{\frac{9}{2}}\frac{l}{T} \tag{18}$$

Least energy E_2 gives least speed V_2; similarly, we get

$$V_2 = \sqrt{\frac{1}{2}}\frac{l}{T} \tag{19}$$

2. Determine the magnetic-field boundary

In the shown Cartesian field frame, entering at O,

